

ESE 499 Undergraduate Research –Summer 2026

**Simulation**

**Comparative Analysis of Environment-Aware Resilient Wireless  
Communication Links in Disaster Scenarios Using SDR**

Name: Sohee Kim

Student ID: 117643907

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Email: [sohee.kim.3@stonybrook.edu](mailto:sohee.kim.3@stonybrook.edu)

Advisor: Prof. Harbans Dhadwal



## 1. Introduction

In the aftermath of severe natural disasters or catastrophic events, conventional macro-cellular communication infrastructure is frequently destroyed or severely compromised. This sudden loss of connectivity creates a critical bottleneck for first responders, search and rescue operations, and overall disaster management. To restore connectivity, emergency response teams increasingly rely on rapid-deployment resilient wireless systems, such as ad-hoc networks or Unmanned Aerial Vehicle (UAV)-assisted relays.

However, the effectiveness and survivability of these emergency communication links are highly dependent on the geographical topology of the disaster zone. The signal propagation characteristics in a dense urban environment, characterized by tall skyscrapers and severe signal scattering (e.g., Manhattan, New York), differ drastically from those in a suburban environment with sparse obstacles and a higher probability of Line-of-Sight (LOS) communication (e.g., Stony Brook).

Before deploying hardware-based testbeds for field verification, it is imperative to establish a mathematical and theoretical baseline to quantify these environmental impacts. Therefore, this paper presents a comparative simulation study to evaluate the performance degradation of wireless communication links in urban versus suburban disaster scenarios. By mathematically modeling Rayleigh and Rician fading channels, this study quantifies the physical limitations of standard communication links and establishes the required Signal-to-Noise Ratio (SNR) thresholds, thereby providing a fundamental baseline for designing environment-aware resilient communication algorithms in the subsequent phases of this research.

## 2. System Model and Simulation Setup

To evaluate the survivability of wireless communication links in disaster scenarios, a baseband equivalent communication system is modeled. The transmitter employs Quadrature Phase Shift Keying (QPSK) modulation to transmit a randomly generated bit stream. The modulated signal is then subjected to two distinct fading channel models representing different geographical topologies:

- **Dense Urban Model (eg. Manhattan):** Assuming the destruction of macro-cellular infrastructure and the presence of dense high-rise obstacles, the environment is characterized by severe multipath propagation with no Line-of-Sight (LOS). This is modeled using a Rayleigh fading channel, where the channel impulse response  $h$  is mathematically modeled as a complex Gaussian random variable.
- **Suburban Model (eg. Stony Brook):** Representing an environment with sparse obstacles where a partial LOS path is highly probable, a Rician fading channel is applied. The Rician K-factor, which denotes the ratio of the signal power in the dominant path to the power in the scattered paths, is set to  $K = 5$ .

Additive White Gaussian Noise (AWGN) is introduced to simulate the thermal noise at the receiver. At the receiving end, assuming perfect Channel State Information (CSI), a Zero-Forcing (ZF) equalizer is applied to compensate for the channel distortion before QPSK demodulation. The simulation evaluates

the Bit Error Rate (BER) across a Signal-to-Noise Ratio (SNR) range of 0 to 30 dB with  $10^5$  transmitted bits per loop.

### 3. Results and Discussion

The simulation results, illustrating the BER performance as a function of SNR for both urban and suburban environments, are presented in Fig. 1

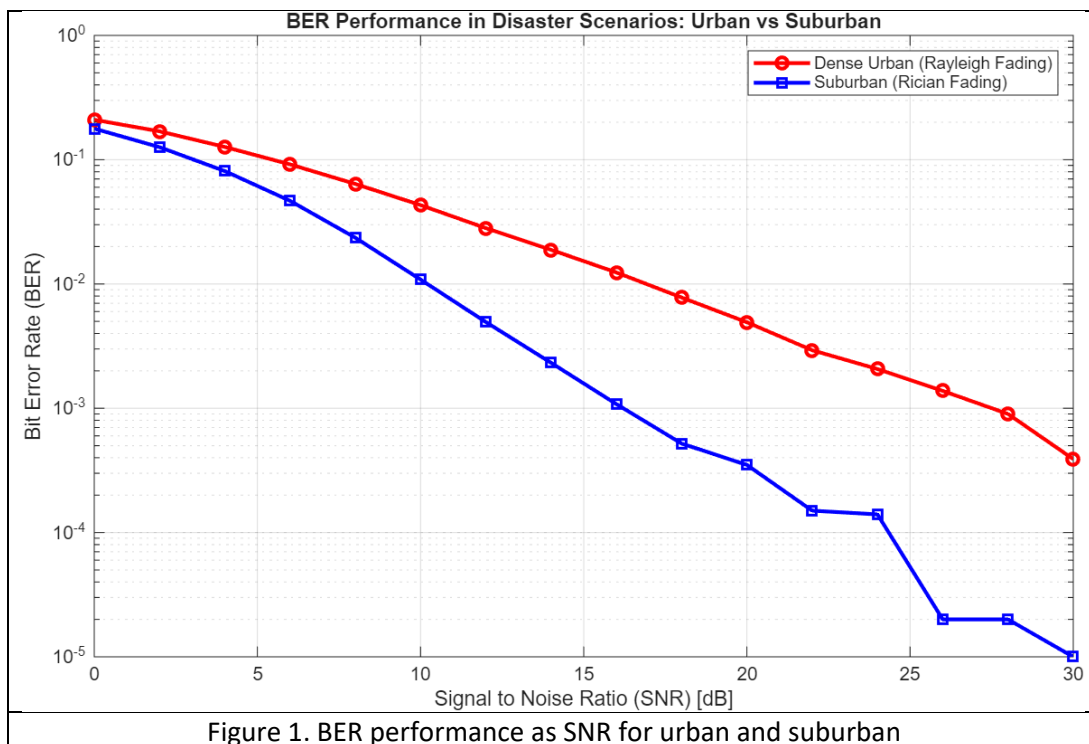


Figure 1. BER performance as SNR for urban and suburban

The empirical curves clearly demonstrate the critical role of the LOS component in maintaining link reliability. To achieve a target BER of  $10^{-3}$ , which is a standard threshold for reliable digital communication, the Rician fading channel (suburban) requires an SNR of approximately 16 dB. In contrast, the Rayleigh fading channel (dense urban) requires an SNR of approximately 28 dB. This reveals a substantial performance gap of over 12 dB.

Consequently, to maintain a resilient communication link in a dense urban disaster scenario, the system must either employ transmission power that is 12 dB higher than in a suburban area, or dynamically switch to more robust modulation and coding schemes (MCS). Furthermore, the step-like artifact observed in the Rician curve at high SNR regimes (above 26 dB) represents the error floor inherently bounded by the finite number of simulated bits ( $10^5$ ).

### 4. Conclusion

This simulation mathematically verifies the severe vulnerability of standard wireless links in urban disaster scenarios. The established 12 dB SNR gap provides a quantitative baseline for the upcoming Hardware-in-the-Loop (HIL) verification using SDR platforms.

## 5. Reference

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- [2] M. Erdelj, E. Natalizio, K. R. Chowdhury, and I. F. Akyildiz, "Help from the Sky: Leveraging UAVs for Disaster Management," *IEEE Pervasive Computing*, vol. 16, no. 1, pp. 24-32, Jan.-Mar. 2017, doi: 10.1109/MPRV.2017.11.
- [3] A. Al-Nahari et al., "Audio Transmission over SDR/Cognitive Radio Using USRP and LabVIEW for Disaster Response," *AIP Conference Proceedings*, vol. 2334, no. 1, 2021, doi: 10.1063/5.0043476.
- [4] A. A. Khuwaja, Y. Chen, N. Zhao, M. -S. Alouini and P. Dobbins, "A Survey of Channel Modeling for UAV Communications," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 4, pp. 2804-2821, Fourthquarter 2018, doi: 10.1109/COMST.2018.2856587.

## 6. Appendix : MATLAB Simulation Code

```
clear;
clc;
close all;

%% 1. System Parameters Setup
M = 4; % Use QPSK modulation (4-QAM)
numBits = 1e5; % Total number of bits to transmit
snr_dB = 0:2:30; % SNR range to test in dB
K_factor = 5; % Rician K-factor for Suburban environment

% Initialize arrays to store BER results
ber_Rayleigh = zeros(1, length(snr_dB)); % BER for Urban (Manhattan)
ber_Rician = zeros(1, length(snr_dB)); % BER for Suburban (Stony Brook)

%% 2. Main Simulation Loop
for i = 1:length(snr_dB)

    % [Transmitter (Tx)]
    % Generate random binary data
    dataIn = randi([0 1], numBits, 1);

    % Convert bits to integers (0~3) for QPSK
    dataSym = bit2int(dataIn, log2(M));
    % QPSK Modulation with phase shift
    txSig = pskmod(dataSym, M, pi/4);

    % [Channel Modeling]
    % Urban: Rayleigh Fading (Severe multipath, No Line-of-Sight)
    h_Rayleigh = (randn(length(txSig), 1) + 1i*randn(length(txSig), 1)) /
sqrt(2);

    % Suburban: Rician Fading (Partial Line-of-Sight exists)
    mean_Rician = sqrt(K_factor / (K_factor + 1));
    std_Rician = sqrt(1 / (2 * (K_factor + 1)));
    h_Rician = mean_Rician + std_Rician * (randn(length(txSig), 1) +
1i*randn(length(txSig), 1));

    % Apply fading to transmitted signals
```



```

rxSig_Rayleigh_fading = txSig .* h_Rayleigh;
rxSig_Rician_fading = txSig .* h_Rician;

% Add Additive White Gaussian Noise (AWGN)
rxSig_Rayleigh = awgn(rxSig_Rayleigh_fading, snr_dB(i), 'measured');
rxSig_Rician = awgn(rxSig_Rician_fading, snr_dB(i), 'measured');

% [Receiver (Rx)]
% Compensate channel distortion assuming perfect Channel State Information (CSI)
eqSig_Rayleigh = rxSig_Rayleigh ./ h_Rayleigh;
eqSig_Rician = rxSig_Rician ./ h_Rician;

% QPSK Demodulation
rxSym_Rayleigh = pskdemod(eqSig_Rayleigh, M, pi/4);
rxSym_Rician = pskdemod(eqSig_Rician, M, pi/4);

% Convert demodulated integers back to bits
dataOut_Rayleigh = int2bit(rxSym_Rayleigh, log2(M));
dataOut_Rician = int2bit(rxSym_Rician, log2(M));

% Calculate and store Bit Error Rate (BER)
[~, ber_Rayleigh(i)] = biterr(dataIn, dataOut_Rayleigh);
[~, ber_Rician(i)] = biterr(dataIn, dataOut_Rician);

end

%% 3. Result Visualization
figure;

% Plot Urban (Rayleigh) results with a red line
semilogy(snr_dB, ber_Rayleigh, 'r-o', 'LineWidth', 2, 'MarkerSize', 6);
hold on;

% Plot Suburban (Rician) results with a blue line
semilogy(snr_dB, ber_Rician, 'b-s', 'LineWidth', 2, 'MarkerSize', 6);
grid on;

% Decorate the plot axes and title
title('BER Performance in Disaster Scenarios: Urban vs Suburban');
xlabel('Signal to Noise Ratio (SNR) [dB]');
ylabel('Bit Error Rate (BER)');
legend('Dense Urban (Rayleigh Fading)', 'Suburban (Rician Fading)');

```